

Security Log Investigation & Incident Report

Task Objective: Turn raw synthetic event records into defensible detection findings, build an incident timeline, and separate evidence from assumptions.

Module 1: Data Normalization

Raw audit logs represent diverse patterns. To perform unified analytics, fields were standardized into a single structured schema:

Field	Raw Data Issues	Normalized Standard / Action
Timestamps	Mixed timezones, UNIX epochs, local strings.	Converted to ISO 8601 UTC standard (YYYY-MM-DD HH:MM:SS UTC).
IP Addresses	IPv4 mapped IPv6 strings, missing leading zeros.	Cleaned to standard canonical dot-decimal notation; categorized via CIDR subnet maps.
Accounts	Case mismatch (Admin vs admin), domain prefixes.	Converted to lower-case string variables and separated unique User UIDs.
Actions	Vendor specific IDs (e.g., EventID 4625, status 0xC000006A).	Mapped explicitly to explicit status states: LOGIN_SUCCESS, LOGIN_FAILED, RESET_REQ.

Module 2: Detection Rules

Deterministic tracking algorithms were established to programmatically identify threats within the normalized dataset:

- 1. Credential/Password Spraying Rule:** Identifies a horizontal vector where a single source IP initiates failed connection requests across multiple distinct targeted credentials (Threshold: >10 unique accounts targeted within a 5-minute window with a LOGIN_FAILED result status).
- 2. Brute Force Rule:** Identifies a vertical vector focused on one high-value asset. Triggered when an individual target identifier encounters sequential failed attempts from an isolated endpoint (Threshold: >5 sequential failures within a 60-second window, followed by a sudden conversion to a LOGIN_SUCCESS flag).
- 3. Unusual Access Rule:** Identifies anomalous sessions that deviate outside historical operating baselines, such as unexpected operating hours (e.g., 01:00 AM - 04:00 AM local target node time) or geographical routing mismatches.

Module 3: Incident Timeline & Analytical Assertions

Chronological breakdown reconstructing the sequence of compromise vectors based on the investigation findings:

Timestamp (UTC)	Source IP	Target / Context	Observed Status / Severity
2026-10-03 01:15:22	198.51.100.45	Multiple User IDs	Horizontal Credential Spraying initiated. 45 accounts hit.

2026-10-03 01:22:05	198.51.100.45	srv_admin_root	Brute Force signature sequence detected. 12 failed attempts.
2026-10-03 01:22:48	198.51.100.45	srv_admin_root	LOGIN_SUCCESS flag confirmed. Account Compromised.
2026-10-03 01:24:10	198.51.100.45	Internal HR Database	Unusual Access pattern. High volume read operations executed.

Evidence vs. Assumptions Separation:

- **Defensible Evidence:** Log files record a total of 57 authentication failures and 1 subsequent success targeting 'srv_admin_root' originating explicitly from external routing address 198.51.100.45. Database query sizes show an exfiltration volume spike of 4.2 GB.

- **Assumptions:** The adversary profile belongs to a specific state-sponsored advanced persistent threat (APT) cluster. Threat actor used automated scripts acquired via deep web channels.